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Cohort profile: a prenatal birth cohort study of intergenerational risk and resilience after conflict and forced displacement

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ABSTRACT

Purpose As of 2024, 123.2 million people had been forcibly displaced as a result of persecution, armed conflict or climate-related catastrophes, and these numbers are predicted to rise. There is a growing awareness of possible intergenerational effects of trauma on life-course health and well-being, however few studies have followed individuals longitudinally starting prenatally. This paper describes the first large prenatal birth cohort study in a refugee context in a lower middle-income country. This study aims to investigate the potential lifespan health and developmental implications of being born into a protracted humanitarian context, and what factors can buffer from the adversity posed by conflict and displacement.

Participants We outline our approach of recruiting, consenting and gathering data from pregnant Rohingya refugee and host community women (N=2888; 80% Rohingya) over the course of 12 months in Cox's Bazar, Bangladesh.

Findings to date A fifth wave of data collection, when children were 6 months old, was completed in April 2025. Rohingya women were substantially less literate; were marrying and having children at slightly younger ages, were more likely to live in crowded, resource-limited households and exhibited higher rates of clinically significant post-traumatic stress disorder and anxiety than host community women.

Future plans There is a critical need for research in displaced populations in order to elucidate potentially lasting transgenerational impacts of experiencing conflict and displacement trauma, and the prenatal and postnatal factors that support health and development across the life span. The next follow-up is planned when the children turn 36 months of age (starting March 2026).

STRENGTHS AND LIMITATIONS OF THIS STUDY

⇒ The breadth and depth of methods used to capture processes and mechanisms at multiple levels of the developmental ecosystem enables us to dive deep into questions of how conflict and displacement trauma affects the health and development across generations.

⇒ The inclusion of a number of neurobiological measures will allow us to investigate the underlying mechanisms of how trauma is transmitted, prenatally, to offspring, and the varied sources of risk and resilience that can buffer or exacerbate biologically embedded risk.

⇒ The large sample size provides ample opportunity to investigate early developmental trajectories and the heterogeneity in outcomes.

⇒ A variety of adaptations to data collection strategies needed to be made to accommodate changes in the political and security situation in the camps and severe weather-related and seasonal occurrences.

⇒ The contextual realities restricted our data collection to one, nonetheless the largest, cluster of refugee settlements setting limits to the number of pregnant women we were able to identify, and thus precluded us from drawing a random sample of pregnant women from the household listing.

INTRODUCTION

By the end of 2024, approximately 123.2 million people had been forcibly displaced as a result of persecution, armed conflict or other forms of human rights violations. The overwhelming majority (73%) originate from and reside in low and middle-income countries (LMICs).[1] At least 40% of those forcibly displaced are estimated to be children ages 0–17 years. Between 2018 and 2024, 2.3 million children were estimated to have been born as refugees. These numbers are only expected to rise in the future as armed conflicts threaten to displace increasing numbers of people with likely implications for

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life-course health and well-being of children being born in displacement. In addition, large influxes of refugees into neighbouring countries—many of which are already struggling to meet the needs of their own populations—can place substantial strain on under-resourced health, housing and education systems, while also intensifying pressure on labour markets and limited natural resources.

Trauma, mental health and intergenerational transmission

Individuals experiencing conflict and forced displacement often suffer significant traumatic and stressful life events before, during and after displacement, which are associated with high rates of depression, anxiety and post-traumatic stress disorder (PTSD).[2 3] There is a growing body of evidence that trauma and trauma-related mental health problems, both before and during pregnancy, can be passed on across generations and lead to worse outcomes for pregnancy, birth and life course health and development of offspring.[4–9] This is often referred to as *fetal origins of health and disease* or the Barker hypothesis.[10]

Variability and sources of risk and resilience

There is significant heterogeneity in the effects of experiencing conflict and displacement on the mental health of mothers-to-be and fathers-to-be as well as on pregnancy, birth and offspring outcomes. This may stem from genetic predispositions to physical and mental health pathologies, levels of exposure to and types of traumatic events, development of subtypes of PTSD as well as a range of other individual, social and contextual factors that may moderate these associations.[8 9 11 12] Multimethod longitudinal studies are needed to better understand the factors leading to this heterogeneity in outcomes.

Longitudinal studies of human development have identified protective processes and sources of resilience that have informed practice and policy to reduce the effects of early risk and vulnerability. The powerful long-term influences of parental mental health and well-being, early attachment relationships, responsive parenting, quality of early care and learning programmes and early economic factors in households were all shown to have large effects on children's development and thriving, which made possible a new era of science-based human development policy in the 21st century.[13–15] While there have been some notable birth cohort studies from LMICs, most long-term longitudinal studies do not start prenatally. The dearth of longitudinal cohort studies that begin in the prenatal period with forcibly displaced populations residing in LMICs is even more stark.

The absence of large longitudinal studies of human development in humanitarian contexts signifies a large gap in the literature in light of the ever-growing number of forcibly displaced populations. Two large initiatives on early childhood development in humanitarian contexts (Ahlan Simsim and Play to Learn) have begun to address the lack of studies evaluating the impacts of early life services provided by non-governmental organizations (NGOs),[16] yet longitudinal studies of significant length and sample size starting prenatally and focused on reproductive health and/or early childhood development with refugee or asylum-seeking populations are few and far between, and we are not aware of any conducted in a humanitarian context in an LMIC. In fact, in two recent systematic reviews of reproductive health in refugee and asylum seekers,[17 18] all were from high-income countries, with one exception, a cross-sectional study in Pakistan of women giving birth in public hospitals.[19] Displaced populations that end up in high-income countries are likely less vulnerable for a variety of reasons, thus limiting the conclusions to be drawn for populations like the Rohingya refugees in Bangladesh.

This study

In light of these gaps in the evidence base, we set out to launch the first-ever large-scale, interdisciplinary, multimethod prenatal birth cohort study in one of the most challenging humanitarian contexts in the world: the Rohingya refugee camps and surrounding host communities in Cox's Bazar, Bangladesh.

The present study aims to investigate (1) the effects of being conceived, gestated, born and raised in contexts of conflict and displacement on health and development; (2) the mechanisms underlying the transmission of prepregnancy experiences of conflict and displacement trauma to offspring and (3) the sources of heterogeneity in these associations in light of a number of risk and protective factors at multiple levels of the bioecosystem.

COHORT DESCRIPTION

Study design

The intergenerational risk and resilience of Rohingya in discplacement (iRRRd) study is an ongoing prospective prenatal birth and early years cohort study. To answer the overarching research questions, we intentionally designed the study to capture factors and processes at multiple levels of the bioecosystem using Bronfenbrenner and Morris' bioecological systems framework.[20] Bioecological systems theory situates the developing individual with its unique biology within its social, physical and symbolic environment. Development is driven by proximal processes, dynamic interactions with individuals and objects (eg, toys) in their microsystem. These microsystem interactions are influenced by factors in the exosystem, for instance, the father's workplace, with which the child does not directly interact (eg, hard physical labour may affect a father's energy to play with the child at the end of the day). All of these settings and the relationships that are cultivated are influenced by larger macrosystem factors, including cultural norms (eg, around parenting) and political and weather events out of the individual's immediate control yet powerful influences on the lived experiences of the developing individual. Finally, the chronosystem represents historical time and chronological age of the developing child rooted in the notion that the systems change as the child develops and interacts

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with an increasing number of individuals and settings, that there are more developmentally sensitive periods (or windows of opportunity), and that risk and protective factors change over time. The interdisciplinary approach underlies a multimethods strategy combining biological with cultural processes of human development and functioning.

We include a subsample drawn from the host communities surrounding the camps. Some careful comparisons can be made as the host community did not experience conflict and displacement of the kind the Rohingya did. However, despite shared cultural and contextual experiences and exposures, the host community is not a perfect comparison. Nonetheless, studying the host community and their experiences has its own merit.

Between February 2023 and March 2024, we enrolled 2888 pregnant women (2322 Rohingya living in camps and 566 from the local host community) at various stages of their pregnancy. Those recruited in their first and second trimesters received an additional pregnancy visit during their third trimester. The birth follow-up consisted of two visits, one within 72 hours of birth, which was kept as short as possible, and one at 28 days postpartum. Participants were followed up again at 6 months postpartum.

Based on the bioecological framing, we included a subsample of husbands and existing siblings aged 36–59 months. Visits with husbands and siblings were scheduled separately, if possible, 2 to 4 weeks after the visit with the pregnant woman, and later mother and target child. Surveys and assessments relating to husbands, target child and siblings will be detailed elsewhere.

We implemented a comprehensive set of measures of potential prenatal and postnatal exposures to risk and protective factors in relation to our main outcomes of interest. In addition, extensive qualitative work and community engagement activities accompany the large-scale data collection activities. Findings are constantly triangulated, forming the basis for interpretation and future research questions.

Statistical power

The main research questions and outcomes of interest pertain to the target child: infant birth outcomes and neurocognitive development at the age of 6 months. The sample size was decided by power calculations for our primary child outcomes.

We used the following equation to calculate sample size using the child development outcomes from the Bayley-IV (see table 1) collected in our pilot (N=300) for both cognitive and receptive language:

$$n = 1 + \rho(m-1) * \frac{Z^2_{\alpha/2}\sigma^2}{E^2}$$

Based on our pilot findings for cognitive and receptive language raw scores, our sample size is 2800, with a 95% CI, 5% significance level and 20.52 SD. We have considered intracluster correlation 0.001 with 30% attrition (refusal, failure to follow-up, miscarriage, stillbirth and child death).

Table 1 Parameters for statistical power calculations to determine required sample size using cognitive and receptive language scores from the Bayley-IV pilot data (N=300)

Indicators	Cognitive	Receptive language
σ=SD deviation	16.03	20.52
Z=z-score corresponding to the level of significance	1.96	1.96
E=desired precision	0.741	0.929
ρ=intracluster correlation	0.001	0.001
m=number of individuals in each cluster	150	150
DF=design effect	1.149	1.149
n=required minimum sample size	2065.68	2153.55
Attrition (30%)	620	646
Total	**2685**	**2800**

More generally, we wanted to make sure that our sample size allowed for important subgroup analyses and advanced and complex statistical modelling often used in the developmental and health science such as latent class analyses and multilevel structural equation modelling of nested data. We reviewed the literature for landmark longitudinal developmental birth cohort studies, for instance, the Dunedin Multidisciplinary Health and Development Study with an initial sample size of 1037 (in Dunedin, New Zealand)[21] as well as landmark longitudinal epidemiological cohort studies, such as the COHORTS collaboration (Consortium On Health Outcomes Research in Transitioning Societies consisting of cohort studies in Brazil, South Africa, Philippines, Guatemala and India), which have provided rich information on developmental processes with initial sample sizes between 2392 (Pelotas, Brazil) and 8181 (Delhi, India).[22]

Location and setting: the Rohingya context in Cox's Bazar

The cohort for this study is based in the most southeastern part of Bangladesh, in Cox's Bazar district, bordering Myanmar. Since the 1970s, this region of Bangladesh has hosted several waves of Rohingya refugees who have fled discrimination, persecution and violence in Myanmar. The most recent and largest displacement began in August 2017, when the number of Rohingya refugees surged from around 250 000 to over 900 000. Most settled around two existing registered refugee camps (Kutupalong and Nayapara), leading to a massive expansion of makeshift structures subdivided into a total of 32 new administrative units (referred to as camps). With an average population density of 50 299 persons per km^2 and over 600 000 inhabitants, one of the largest clusters of camps, conjoined with one of the registered camps (Kutupalong), is now the

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world's largest refugee camp. Figure 1 provides a visual representation of the geographic location of camps.

Within the camps, most families are almost completely dependent on humanitarian assistance and resources provided by non-governmental and governmental organisations. In some families, a male family member may have managed to find work in the adjoining communities, but this is not officially permitted, and work is hard to come by. To date, Rohingya legally are not allowed to work nor go to school as they have not been officially given 'refugee' status under Bangladeshi legal definition. There are also some opportunities for stipended volunteer engagement with organisations working within the camps; but these are few and far between and limited in scope.

The camps were established between and alongside existing communities (ie, the host community) up and down hilly terrain previously covered by lush forests. The host community relies on traditional livelihoods in agriculture, fishing and small-scale trading. Even before the large-scale Rohingya influx in 2017, this community faced significant socioeconomic challenges, including high poverty rates, limited access to quality education and inadequate infrastructure. The rapid expansion of refugee settlements has further strained local resources, especially affecting land use, water sources and the local labour market. At the same time, the humanitarian presence has introduced new economic opportunities. However, environmental degradation, notably deforestation and groundwater depletion, has emerged as a pressing issue, disrupting traditional agricultural practices and natural resource-based livelihoods. Given the geographic proximity to the camps, Rohingya and host communities are affected by many of the same challenging circumstances, including weather-related disasters and crime.

Sampling strategy and recruitment

The camps, and entry permissions to them, are regulated primarily by the Refugee Response and Repatriation Commissioner (RRRC), a specialised Bangladesh government agency. Any plans and corresponding research instruments are reviewed before access to the camps is granted. Each camp is then overseen by a Camp in Charge, a Bangladeshi government appointee, whose permission also needs to be sought. Within the camps, *Majhis*, male community leaders, serve as block-level intermediaries between the community and camp authorities, NGOs and other actors. Good relationships with *Majhis* are critical for community engagement and trust building, can facilitate recruitment and help prevent misinformation and rumours. Our senior research field managers were instrumental in maintaining good relationships with all levels of camp governance, which has enabled smooth operations and high rates of participation and sample retention.

We identified 10 camps from the cluster closest to Ukiah and adjacent to the Kutupalong camp. Female Rohingya volunteers, most of whom lived in Kutupalong camp, were hired to support the recruitment process and conduct the first birth visit. Recruitment took place over a 12-month period. This was done for both logistical and scientific reasons: managing field logistics in this complex and challenging context and capturing seasonal influences and exogenous shocks (weather, fires, etc). The inclusion criterion was pregnancy at the time of recruitment; there were no exclusion criteria. We set out to recruit from 10 camps continuously across 12 months that were selected to capture variability in terrain and humanitarian infrastructure (eg, availability of health clinics and maternity facilities). However, logistical and security challenges meant adapting our recruitment strategy. It proved logistically too challenging to recruit in all 10 camps simultaneously, so the recruitment was focused on three camps at a time. Recruitment data collection was by far the most labour-intensive wave given the sheer number of families to process per week, whereas consecutive waves would spread out over 18–20 months (rather than 12) due to pregnant women having been recruited at any stage of pregnancy. We also modified the selection of camps to correspond with where our Rohingya volunteers lived as traveling between camps was logistically very challenging, especially for the Rohingya volunteers whose movement is restricted by camp regulations. We had to interrupt recruitment in one camp due to security concerns, but restarted data collection once the situation was stable again, but ultimately ended up with eight camps.

A flowchart (Figure 2) lays out details of the sampling and recruitment process and details for sample retention in the subsequent, completed waves of data collection. We conducted a household listing of pregnant women. Female Rohingya volunteers listed a total of 5068 households in the camps and female enumerators from the local host community listed 621 households in the host community as having a pregnant woman. The sample was stratified by age (a target of 60% below the age of 19 to ensure representation of first-time pregnant and primiparous women) and aimed for proportional representation by camp size. However, it became increasingly difficult to identify pregnant women under the age of 19 in the 10 camps, so we adjusted the target downward, resulting in approximately 41% of pregnant women under the age of 19. From the remainder of the listing, we then prioritised pregnant women based on proximity from the under-19 pregnant women's households until the target number for that camp was reached. If a new pregnant woman was identified, she was added to the listing. Given low refusal rates (five cases; reason was no time; three of them were revisited on a different day) of pregnant women, we are confident that our sample is likely to be unbiased and reflective of the community at large.

Data collection procedures

We provided participants a small monetary incentive, and fortified biscuits in homes where other children were present. The monetary incentive was determined by the Office of RRRC. For pregnant women, it is 150 taka (approximately US$1). Aside from the incentives, we also provided assessments that are of utility to participants

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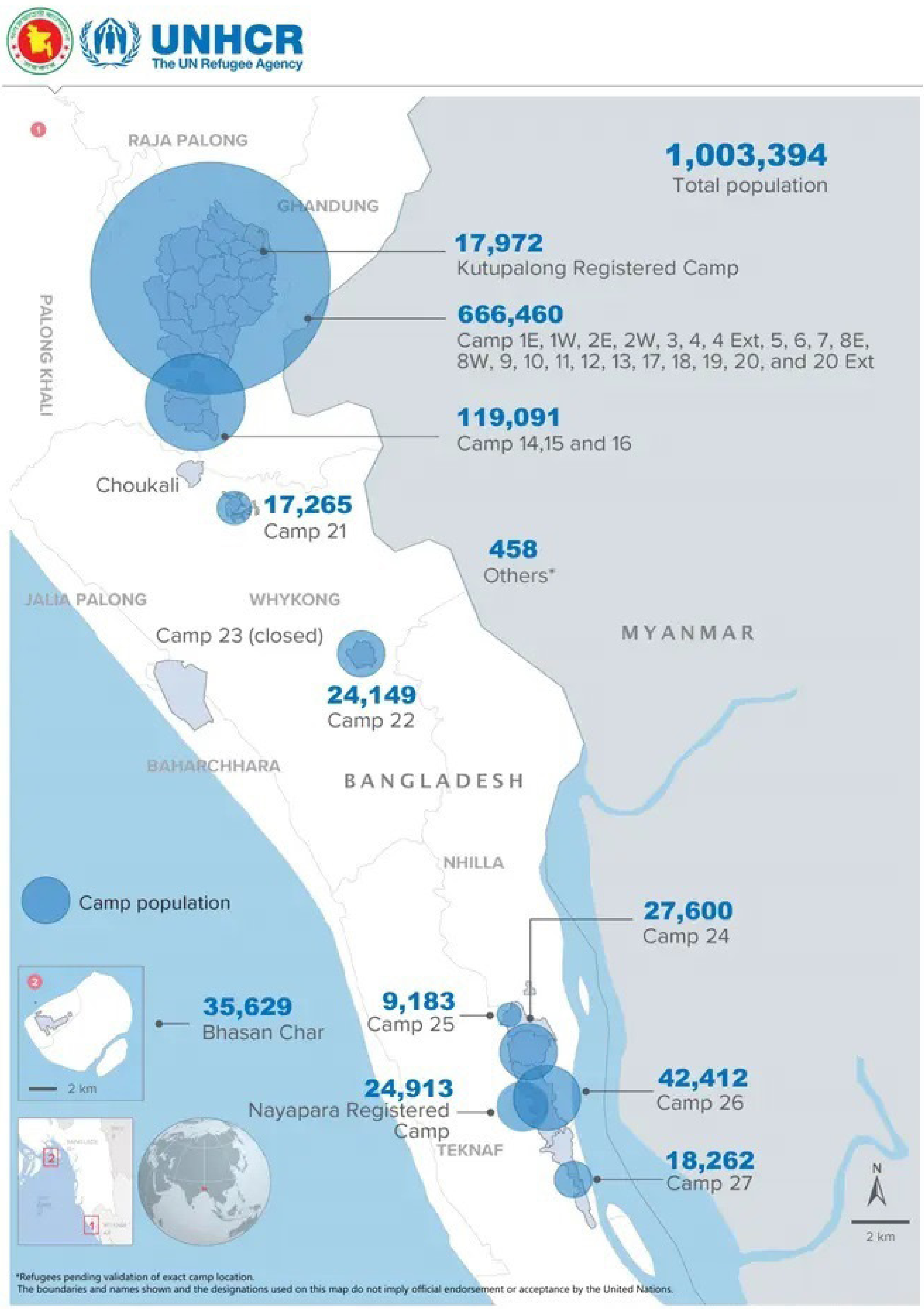

Figure 1 Map of Rohingya refugee response in Bangladesh indicating Rohingya population by location (as of 30 September 2024). Source: creation date 30 September 2024, GoB-UNHCR joint Registration Exercise; https://reliefweb.int/map/bangladesh/rohingya-refugee-responsebangladesh-rohingya-population-location-30-september-2024

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(blood pressure, haemoglobin, glucose). If these readings were outside the normal ranges, they were advised to seek medical attention. In addition, we hired an experienced protection officer (child and social protection) who spoke Rohingya and Chittagonian (local language) and knows the camp context well; her job was to provide referrals and guidance when participants asked for additional support. Additionally, two of our senior field staff were physicians and able to provide additional support and advice in cases requiring medical attention.

Linguistic particulars of working with Rohingya require enumerators to be from the area of Bangladesh where the local language (Chittagonian) has a significant although not complete overlap with the Rohingya language. In addition, all of our survey enumerators had previous experience working in the camps (for details on conducting high-quality research in this context, please see Goodfriend *et al*.[23]

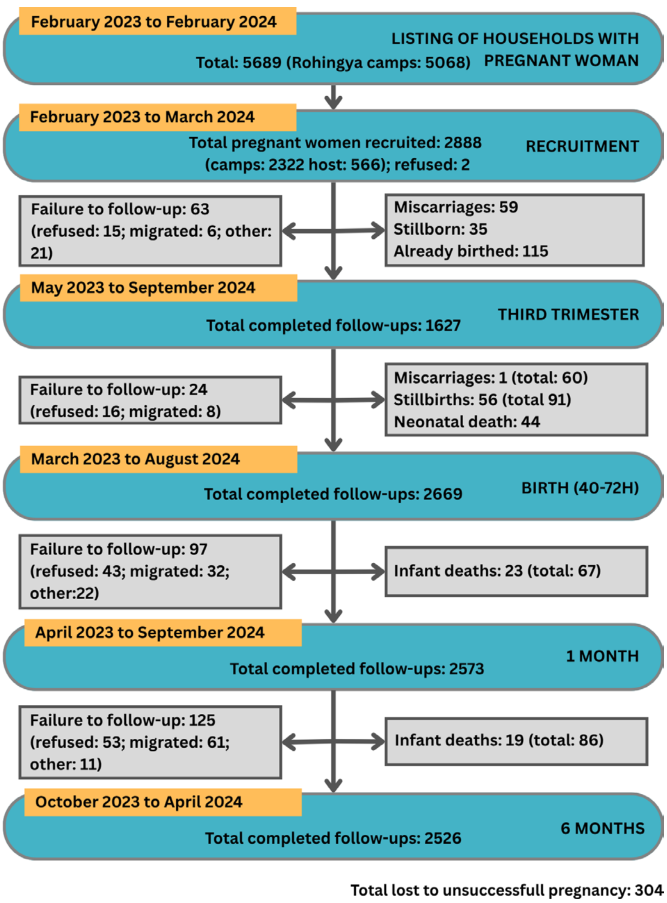

Figure 2 Flowchart of sample recruitment and follow-up. Notes: The 'successfully completed follow-ups' refer to participants found where data collection was completed; this number equals the full sample (2888) minus 'failure to follow-up', minus the totals of pregnancies and babies lost up to that timepoint; this does not hold for the third trimester follow-up as not all women received a third trimester visit. The table only reports on women/mothers successfully followed up with and not how many neonates/infants were successfully assessed alongside the mothers. Some households returned at a later time point; one woman miscarried and got pregnant again between recruitment and third trimester follow-up; some individuals absent at one wave were available again at the next one.

The collection of biomedical measures and samples, as well as anthropometric and other direct assessments and observations, requires different skill sets and/or intensive specialised training. Thus, depending on the activities scheduled for a visit, the primary survey enumerator, who often also functioned as a translator and relationship manager, was accompanied by medical technicians (MTs). In general, survey enumerators conducted one recruitment survey per day; MTs visited two or three households where survey enumerators were conducting recruitment surveys, to collect biomedical and neurobiological data. Given the cultural specifics of this context, any interactions with female participants were conducted by female data collectors and all interactions with male participants were conducted by male data collectors.

Female enumerators visited households from the listing to provide details on the study, answer any questions the potential participant and/or other family members might have and explain any potential implications of participating in the study. Formal consent procedures were conducted after answering all questions and concerns satisfactorily. Consent was taken verbally due to high rates of illiteracy, especially among the female Rohingya population.

After obtaining consent, data collection activities commenced. Given the sensitivity of certain questions—such as those related to traumatic experiences, mental health and domestic relationships—ensuring privacy during interviews was a key concern. Data collectors made efforts to create private spaces and paused interviews when privacy was compromised. This was particularly challenging in the camps, where small, crowded dwellings and frequent foot traffic made uninterrupted conversations difficult. To navigate these constraints, data collectors employed creative and context-sensitive strategies. The most effective approach to establishing privacy involved building trust through the inclusion of family or community members in explaining the research, and securing the support of *majhis* and other community leaders.

Patient and public involvement

Early focus group discussions with community leaders and members of the community reflecting the targeted study population informed the research design and allowed us to gauge priorities of the community. Specifically, it informed the selection of biological measures with regards to cultural appropriateness, and the community's desire for meaningful information with regards to their health and well-being. We engaged community members and key informants in the adaptation and development of culturally grounded measures. Ongoing focus group discussions are held to get community reactions to early results, or to gather additional information when encountering surprising or conflicting results.

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Measures

This longitudinal study builds on an extensive mixed-methods pilot study conducted to (1) better understand the sociocultural and contextual specifics of the study population; (2) develop, adapt, test and validate a variety of measures and methods; and (3) test and strengthen field infrastructure and strategies given this extremely difficult context. The pilot phase focused exclusively on the Rohingya population, as many of the instruments had already been used in Bangladesh with the host community. New or untested measures were carefully reviewed by the icddr,b team and refined through workshops with enumerators to ensure accurate translation and contextual relevance. Although new instruments were also translated into Bangla for the host community, interviews were ultimately conducted in the local dialect, Chittagonian. Our enumerators—native speakers of Bangla and Chittagonian, and largely fluent in Rohingya where it diverged from Chittagonian—played a key role in this process. Through extensive workshoping, we developed satisfactory versions for each population. Where ambiguity remained, Rohingya terms were added directly to the survey script.

Due to pandemic-related shutdowns, the pilot study spanned from mid-2019 through late 2022 and included (a) 92 in-depth semistructured interviews with pregnant women, husbands of pregnant women and adolescent boys and girls; (b) surveys with a total of 727 pregnant and parenting women and parenting men; (c) neurodevelopmental assessments with 597 children aged 1 month to 7 years; and (d) six focus group discussions with women, men and community leaders exploring the acceptability and feasibility of a range of approaches to biomarker collection.[24] In addition, we conducted a mixed-methods study to develop, adapt and test measures related to mental health and functioning, caregiving and socioeconomic status specific to the sociocultural context.[25] Based on this pilot study, we are confident that our measures and methods are culturally and contextually appropriate.

In the following descriptions of measures, we do not report on some of the specifics, including the number of items in each measure or the specific sections from the measures we used. For one, we adapted measures to varying degrees before and after piloting based on considerations of appropriateness, acceptability, feasibility and meaningfulness. Some items were also dropped based on (a) psychometric considerations; (b) attempts to reduce the length of surveys; and/or (c) the request of the RRRC. More details on measure adaptation and validation are found in Goodfriend *et al,* Trang *et al and Wuermli et al.*[23 25 26]

In this cohort profile paper, we describe measures and assessments implemented at recruitment with pregnant women. A complete list of measures and assessments implemented at follow-up visits and with the target child once born is found in online supplemental table S1 of the supplemental materials. As of April 2025, the 6-month follow-up has been concluded, and we are in the process of cleaning and analysing that data.

Demographics, household structure and resources

We assessed pregnant women's demographics, socioeconomic status and household structure using standard questions, access to resources using the Perceived Refugee Index (PREI),[27] household size and assets, and questions tailored to document their migration history. The PREI is a multidimensional measure and has been developed specifically to assess the quality of refugee environments, as there are currently no comparable measures of refugee settings.[27] The questions, however, apply to low-resource non-refugee settings also.

Social support, agency and relationships

We developed a study-specific set of items to assess pregnant women's social support, their agency and ability to leave the house on their own or in company of a family member and drew on sections of the Responsible Engaged and Loving Fathers[28] survey to assess pregnant women's intimate partner support, violence and engagement, specifically section (2) *Background on Partner Decision-making and Information about Family Planning*; and section (3) *Relationship with Partner, Communication, and Attitudes Toward Intimate Partner Violence.*

Health and pregnancy

Women responded to a number of questions related to (1) general health, injury and illnesses; (2) perinatal health and pregnancy; and (3) access to and take up of available prenatal services, and the procedures and tests they recalled receiving.[29] Details on antenatal health care behaviors are provided in Simon *et al.*[30]

Biomedical and anthropometrics

Pregnant women received a few select direct assessments of general indicators of health administered by a MT with support from the survey enumerator if needed. Pregnant women were weighed and measured for their height. Reliability was assessed by repeat measurements: two if the second measure was equivalent to the first, three if the second one was different.

The MT further took blood pressure and measured glucose and haemoglobin by finger prick and point-of-care devices. Blood pressure was measured using the auscultatory method with the manual sphygmomanometer and stethoscope. For haemoglobin, we used the Hb 801 device by HemoCue; for glucose, we used the Accu Chek Instant Sugar. These indicators were incorporated not just for scientific purposes, but to provide our participants with information relevant to their pregnancy health, with recommendations to seek a follow-up exam at a local health clinic should values fall outside of the normal range.

Mental health and functioning

We assessed mental health and well-being using a number of modified scales. For depression and anxiety, we used a slightly modified 25-item version of the Hopkins Symptoms

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Checklist (HSCL-25)[31] supplemented with 15 items from a mixed-methods study investigating cultural concepts of distress[25]; the expanded measure is called the *Rohingya Demaki (mental health) Measure*. Similarly, we developed a gender-specific measure of functioning, the *Rohingya Lorasora (functioning) Measure*, reflecting the (lack of) difficulty that Rohingya experienced in engaging in culturally appropriate daily chores and activities. For PTSD, we used an expanded version (to capture all DSM-5 criteria) of the Harvard Trauma Questionnaire section 4 (HTQ-4).[32] We assessed quality of sleep with the Pittsburgh Sleep Quality Index.[33] Pregnancy-specific anxiety was captured with adapted items from the DHS.

Trauma exposure

We assessed pregnant women's childhood trauma exposure and war/migration-related trauma at recruitment, using the Adverse Childhood Experiences International Questionnaire[34 35] and the HTQ-1.[36]

Neurobiological measures and specimens

We included assessments of maternal parasympathetic and sympathetic functioning via heart rate variability (HRV) and electrodermal activity (EDA), respectively.[37 38] We use two field-friendly devices, Firstbeat Bodyguard3 to acquire ECG for HRV, and eSense for EDA. Studies have shown associations of maternal HRV prenatally[39] and postnatally[40 41] with offspring development.

We collected a saliva sample which was then aliquoted into three separate cryovials. These samples underwent a cold change management process and were ultimately stored at −80°C until assayed. Saliva will be assayed for proteins and metabolites (eg, inflammatory markers, steroid hormones, oxylipins) and used to investigate potential mechanisms underlying the link between trauma experiences and mental health problems, and trauma and mental health associations with offspring development.[42–45] We also collected buccal swabs using iSWAB-DNA-250 Collection Kit from Mawi DNA Technologies LLC for epi/genomics.[46–48]

Measures used in follow-ups

It exceeds the scope of this paper to describe in detail additional measures that were collected in follow-up waves. Table 2 does list all the measures collected at each of the completed follow-up waves for both mother and child. Broadly speaking, we do not repeat measures that capture past experiences that are not subjected to change (eg, trauma experiences, reproductive health when not pregnant), but we do reask all relevant measures that can change over time (eg, mental health). In addition, we collect intensive data from direct assessments and caregiver report on the baby at three timepoints.

Description of the cohort of pregnant women

Table 2 shows the details on the characteristics of study participants (see online supplemental table S2 in the supplemental materials for complete summary statistics, including broken down by Rohingya and host community). There is generally very little missingness in these variables. The 166 missing observations for PTSD were due to a programming error in our survey which we noticed a couple of weeks into recruitment. Any differences between Rohingya and host would stem from rate of recruitment rather than refusals (5.6% Rohingya; 6.4% host). Nonetheless, the difference is not statistically significant (p=0.49). The missing observations for husband's age are from Rohingya only (0.006% Rohingya; 0% host) and possibly due to higher rates of illiteracy compared with the host community, though the difference is only marginally statistically significant (p=0.06). At recruitment, pregnant women had a mean age of 21.20 years, almost 41% were between the ages of 12–18 years old. Rohingya and host community women did not differ significantly in average age, although there were more Rohingya women in the youngest age group, under 16, and more host community women in the 19–34-year age group. The husbands of Rohingya women were on average 1.3 years younger than their host community counterparts. Women were married on average before the age of 16, with Rohingya women married 0.88 years earlier than host community women. Although literacy is a complicated subject in the Rohingya context, we ascertained that 68% of Rohingya women in our sample could not read at all, in any language, whereas only 8% of host community women were illiterate. Approximately 56% of women were recruited in their first or second trimester. For 29%, this constituted their first pregnancy, and for 35%, this pregnancy, if viable, would be their first child. There were significant differences between Rohingya (26% first pregnancy, 32% first child) and host community women (41% first pregnancy, 50% first child) in these characteristics.

The vast majority of Rohingya (91%) came to Bangladesh fleeing targeted, systemic violence in Myanmar in 2017/2018. As expected, Rohingya women had experienced significantly more conflict and displacement trauma events than their host community counterparts. Correspondingly, 12% of Rohingya compared with 5% of host community women scored above the cut-off for clinically significant PTSD symptoms. For depression and anxiety, on the other hand, Rohingya and host community women showed similar symptom scores, with 44% and 61% overall scoring above the cut-off for clinically significant distress, or symptoms of depression and anxiety; only clinically significant anxiety was significantly more prevalent in Rohingya (63%) than host community women (54%). These statistics stand in contrast to previous work with the Rohingya in Bangladesh where researchers found that 4.9% scored above threshold for PTSD, and 30% scoring above the cut-off for depression.[49] These discrepancies could be due to the nature of the sample (ours being younger and female only), and in the case of depression, the scale we used was HSCL-25, whereas the prior study used the PHQ-9.

On average, participants lived in five-person households and lived in two room dwellings. However, and

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Table 2 Sample descriptives (reported by pregnant women at recruitment)

	N	Miss	Mean/%	(SD)	95% CI	Min	Max
Age	2888	0	21.20	(5.13)	(21 to 21.38)	12	40
Younger than 16 years			6.9	–	(0.06 to 0.08)	0	1
16–18 years			34	–	(0.32 to 0.36)	0	1
19–34 years			57	–	(0.55 to 0.59)	0	1
35 years and older			1	–	(0.008 to 0.02)	0	1
Husband's age	2874	14	27.15	(7.25)	(26.88 to 27.41)	14	78
Age when married	2888	0	15.92	(2.17)	(15.84 to 16.00)	10	28
Cannot read in any language	2888	0	56	–	(0.54 to 0.58)	0	1
First pregnancy	2888	0	29	–	(0.27 to 0.30)	0	1
First child	2888	0	35	–	(0.34 to 0.37)	0	1
Year of arrival in BD (Rohingya only)							
Before 2017							
Between 2017 and 2018							
After 2018							
Conflict and displacement trauma events	2888	0	0.14	(0.001)	(0.13 to 0.14)	0	0.47
PTSD symptoms score	2722	166	1.77	(0.53)	(1.75 to 1.79)	1	3.79
Above diagnostic cut-off			10	–	(0.09 to 0.11)	0	1
Depression symptoms score	2888	0	1.74	(0.47)	(1.72 to 1.76)	0	3.86
Above diagnostic cut-off			44	–	(0.43 to 0.46)	0	1
Anxiety symptoms score	2888	0	1.92	(0.46)	(1.90 to 1.94)	0	3.5
Above diagnostic cut-off			61	–	(0.6 to 0.63)	0	1
Resources							
Household size	2888	0	5.05	(2.28)	(4.97 to 5.13)	1	19
Number of rooms in house	2888	0	2.31	(0.97)	(2.28 to 2.35)	0	9
Access to resources (factor score)	2888	0	2.63	(0.35)	(2.62 to 2.64)	1	3

PTSD, post-traumatic stress disorder.

consistent with the higher proportion of first pregnancies in the host community women, more people lived in the homes of the Rohingya than host community families. This was despite Rohingya living in smaller dwellings (fewer rooms) than their host community family counterparts. Finally, with regards to access to resources (scored 1 to 3), host community households reported higher access to resources than Rohingya households.

CHALLENGES AND LIMITATIONS

The context in which this study is being conducted is extraordinarily challenging. Course corrections and minor changes to the fieldwork strategies often were required in response to weather-related calamities (strong rains, cyclones, extreme heat), seasonal spikes of mosquito-borne diseases affecting field staff, power outages leading to interruptions of device charging, surges of crime and violence making certain camps or areas too dangerous to work in, fires in camps and a slowing of recruitment over the 12 months as identifying additional pregnant women became more difficult. Some of these were anticipated but were not preventable (eg, seasonal occurrences), and others were not predictable (eg, crime, fires).

Despite running large and comprehensive pilot studies, some of the logistical challenges associated with the scale of this study could not be fully tested and prepared for, for instance, working in multiple camps simultaneously, and the sheer magnitude of field operations once all activities were running in parallel (ie, birth follow-up visits began while recruitment and prenatal assessments were ongoing). These challenges required some troubleshooting in the early weeks of recruitment, and at the beginning of each new activity in subsequent waves. To account for the possible hiccups in the first couple of weeks of each activity potentially affecting data quality, we oversampled to exceed our original target enrolment.

While we believe our sample is reflective of the population at large, our sampling strategy was not designed to be representative. For logistical and resource reasons, we decided to conduct our study in the largest cluster of camps (near Kutupalong registered camp), and selected 10 camps out of 22 in that cluster. As we wanted to ensure

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that these 10 camps reflected the differences between camps with regards to terrain, size and infrastructure, they were purposively selected. In addition, we had to take into account the amenability of that camp's specific administrative body to approve our research activities. As we wanted to have a large enough subsample of young, primiparous women to conduct subsample investigations for specific research questions related to early and first-time pregnancies and births, this of course skews our sample towards younger pregnant women. Yet, the extremely high rate of expectant mothers agreeing to participate in the study (>99.7% of women approached) offers confidence that there are no other biases in the sample of pregnant women.

CONCLUSIONS AND FUTURE PLANS

Despite these limitations, we remain confident in the quality and rigour of the study, and the immense contributions it will make to our understanding of how conflict and displacement affect early childhood development. In addition to increasing the body of knowledge on the important questions identified above, this study offers excellent potential for translational and applied efforts. Both the data generated in answering these questions, and the new knowledge generated, will be instrumental for informing new programmes and improving existing ones, to better support displaced and vulnerable families in LMICs to ensure the best chances of living healthy and productive lives, from the early prenatal period onwards.

It is worth reiterating a few specific considerations that influenced the design of the study as they address gaps of varying degrees in the literature. Beginning prenatally is of particular importance given the developmental sensitivity during this time regarding stress exposure,[5 9] which is critical given the levels of trauma and stress in this population and context. It also sheds light on a number of factors influencing neurodevelopmental outcomes unrelated to the quality of caregiving.[9]

In line with best practices of human developmental research, we wanted to ensure a truly bioecological and dynamic systems perspective.[20] Assessing and measuring factors and processes at multiple levels of the human ecosystem (from the biological to the sociocultural and environmental) allows for a more comprehensive assessment of dynamic processes and mechanisms that constitute human development, providing insights on sources of risk and resilience outside of the primary caregiver–child dyad. In addition, combining diverse methods and community engagement allows for triangulation of data and verification of our findings, providing a thorough and valid picture of participants and their communities.

There have been virtually no longitudinal studies of early childhood development with forcibly displaced populations that start prenatally and draw from methods and measures used across a diverse range of disciplines, from epidemiology to anthropology. We combined survey research with in-depth semistructured interviews; neurobiological functioning; direct assessments of children's development, anthropometrics and health; observations of children interacting with their caregivers and environment; and publicly available data on extreme weather events, natural disasters and fires that affected large segments of the population within geographic boundaries. We further captured the availability and take-up of health and child development services. Geospatial data on household location in relation to fires and other exogenous events, and services, provide a deeper understanding of contextual factors affecting the lived experiences of the Rohingya population and how they affect human developmental outcomes.

Finally, with regards to collecting neurobiological data, we have made significant investments in setting up a biobank of saliva and buccal cell samples. Our approach[24] has allowed us to identify a number of methods and approaches to tap into stress physiology and neurobiological functioning. Some of the measures and assessments provide on-the-spot data (eg, point-of-care device to measure haemoglobin), others require considerable processing (eg, attaining HRV measures from ECG recordings), and some will require additional and substantial financial and time resources to produce data (e.g., assaying saliva and buccal swabs). Nonetheless, the unique opportunities presented by this study warranted the investment in infrastructure to house a biobank.

Following our prenatal cohort, children for a longer period of time will allow for groundbreaking investigations into protective factors at various stages of their development that can be leveraged for the development of new and improvements of existing programmes to support early childhood development. It will also enable us to investigate longer term effects of maternal and paternal conflict and displacement trauma on the next generation. We have received continued funding from the LEGO Foundation to conduct a follow-up wave of data collection when our prenatal cohort reaches 36 months of age (starting in March 2026). This represents a remarkable opportunity to investigate developmental trajectories using state-of-the-art neurocognitive assessments (including Bayley-IV) at an age where cognitive, emotional and behavioural development are more reliably measured, and where we find stronger predictive power for later developmental and health outcomes.

We are further actively fundraising to realise our plans for the biospecimens currently banked and stored at required temperature to preserve the integrity of the samples. We intend to conduct an intensive deep dive into molecular processes and mechanisms with a particular focus on markers and combinations of markers reflective of stress physiological processes, to elucidate the mechanisms 'under the skin' and invisible to the eye of how trauma affects the health and development of the next generation.

We believe the findings from this study will have wide-ranging implications for how we think about and support children and their families affected by conflict and

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displacement for generations to come. We further posit that this study will not just advance basic science, but by incorporating extensive and cutting-edge methods and perspectives from the fields of molecular biology and medicine, it will push the boundaries of applied developmental and mental health science in LMICs.

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Contributors AW is the guarantor of this study and the manuscript. AW and FT, the Bangladeshi co-principal investigator, co-led the conceptualisation and design of the study. Both have been centrally involved in protocol development and oversight of implementation. All listed co-authors contributed to various aspects of study design, implementation, analysis, or manuscript preparation, and all reviewed and approved the final manuscript. This manuscript represents the first peer-reviewed publication from the study. We have included all individuals who made substantial intellectual contributions to this work, in accordance with ICMJE authorship guidelines. Field teams, including data collectors, are acknowledged in the acknowledgements section. I did not use AI for the content of this manuscript. But I used ChatGPT to get information on Vancouver style formatting. I also asked for details on the Global Burden of Disease 2016 categories of mental health symptom severity and how that would align with official cutoffs of the measures we used, and to expedite some of the literature searches. I'm very new to ChatGPT, so I only just used it for final touches, but I'm starting to see some really good uses for it that really expedite some of the 'manual' labour. RandR: I asked ChatGPT to do some light editing of additions to the manuscript. English is my second language and sometimes I miss commas or misuse a preposition. I also asked Chat for help with the contributorship statement, asked what a guarantor was and what information was expected.

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Data availability statement Data are available upon reasonable request. The iRRRd data are not yet openly available as data collection and processing are still ongoing. However, the project team is working on protocols for proposals to access recruitment data in collaboration with project investigators. Researchers interested in potential collaboration should contact the principal investigators: Drs Alice J. Wuermli (alice.wuermli@nyu.edu) and Fahmida Tofail (ftofail@icddrb.org) to complete a research proposal for evaluation by the iRRRd study leads.

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